

STUDENT RETENTION INTELLIGENCE PLATFORM

An Explainable AI-Based Early Warning, Counseling and Intervention System for Student Retention

SIH Problem Statement: SIH25102 — AI-based drop-out prediction and counseling system

Organization: Government of Rajasthan **Theme:** Smart Automation

This document is the team's complete technical guide: what we are solving, what we are building, why the architecture is split into two phases, what each component does, how the AI loop works, what the team must learn, and how we will test the final system.

Core idea

Predict → Explain → Decide → Intervene → Follow-up → Measure → Re-score

Phase 1: reusable Azure cloud infrastructure foundation. **Phase 2:** the SIH25102 Student Retention Intelligence Platform running on that foundation.

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Team rule: understand the architecture before coding. The goal is not to collect technologies; the goal is to build one coherent system whose pieces work together.

2. The SIH Problem Statement

SIH25102 — AI-based drop-out prediction and counseling system

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The supplied SIH analysis describes a situation in which students may show warning signs before dropping out — such as declining attendance, declining academic performance or fee-related difficulties — while teachers and administrators may lack a systematic way to identify those patterns early and connect them to intervention.

The important word is counseling. A system that only predicts “high risk” is incomplete. The intended value is to help an authorized human understand the risk, inspect the contributing signals and decide what support may be appropriate.

Key challenges identified in the supplied analysis

- Real student data is sensitive/private, so real institutional data may be difficult to obtain.
- False positives and false negatives matter: a wrong prediction can create unnecessary concern or miss a student who needs support.
- Counseling recommendations must be meaningful; the system should not stop at an “at risk” label.

Source-backed baseline solution: public student dataset → EDA → classification models → SHAP explanations → teacher/counselor dashboard → optional alerts.

3. What the Problem Really Means

Imagine a student whose attendance has fallen over several weeks, assignments are increasingly missed and academic performance is declining. All of these values may exist in institutional records, but without an early-warning workflow a teacher still has to notice the pattern manually.

Our system converts those scattered signals into a structured workflow:

Stage	What happens
1. Detect	Estimate current dropout risk from available data.
2. Explain	Show the strongest model contributors for that student.
3. Decide	Prepare a transparent intervention category for counselor review.
4. Intervene	Record the action chosen by the counselor.
5. Follow-up	Schedule and record subsequent observations.
6. Measure	Compare later risk/data with the earlier state.
7. Re-score	Update the student's trajectory when new data arrives.

Core principle: prediction is only the beginning. The useful product is the closed loop around prediction.

4. Our Proposed Solution

Student Retention Intelligence Platform is our proposed implementation of SIH25102. It is a cloud-based, explainable early-warning and intervention platform for authorized counselors/administrators.

Two-phase structure

Phase	Purpose	Major components
Phase 1 — Azure Cloud Foundation	Build a reusable, secure and automated cloud environment.	Terraform, Azure Storage, Linux VM/compute, networking, security baseline, Key Vault, YAML configuration, validation and CI/CD.
Phase 2 — Retention Intelligence Platform	Build the actual SIH application on the Phase 1 foundation.	Data ingestion, risk prediction, SHAP, intervention support, counselor workflow, longitudinal tracking and outcome measurement.

One sentence for the whole project: We are building an explainable student-retention decision-support system, not merely a dropout prediction model.

5. What Makes the Solution Stronger Than a Basic ML Demo

A basic college project can be: upload CSV → train model → show probability → stop. That demonstrates machine learning, but it does not fully address the operational side of SIH25102.

Basic approach	Our intended platform
Risk probability	Risk probability + prioritization
Black-box number	Risk + understandable model contributors
Generic recommendation	Driver-linked intervention category
One-time prediction	Repeated scoring and longitudinal trajectory
Dashboard only	Counselor case/follow-up workflow
No outcome measurement	Intervention history + later risk comparison
Manual cloud setup	Reusable Terraform Azure foundation

Honest positioning: we should not claim that dropout prediction, SHAP or dashboards are individually new. The differentiation is the integrated workflow: reusable cloud foundation + explainable prediction + human decision layer + intervention tracking + longitudinal measurement.

6. Complete System Flow

End-to-end loop

Student data → validation → feature preparation → risk model → SHAP explanation → intervention support → counselor action → follow-up → new data → re-score

Ingest — receive structured student metrics.

Validate — check missing values, types, ranges, duplicates and schema.

Predict — generate probability/risk band using the approved model.

Explain — generate the main contributors to the prediction.

Decide — map the available evidence to a reviewable intervention category.

Intervene — counselor chooses and records an action.

Measure — observe later risk/data changes.

Re-score — update the student's trajectory as new information arrives.

7. Phase 1 — Azure Cloud Infrastructure Foundation

Phase 1 is the engineering foundation. Instead of manually creating Azure resources only for our dropout application, we define the environment as code and organize it into reusable modules.

Phase 1 objective: build a repeatable Azure environment that can be provisioned, reviewed, secured and adapted through Terraform.

The supplied architecture specifies standardized Infrastructure as Code, YAML-driven configuration and security baselines as the main objectives. It also proposes modular repository structure, dynamic provisioning and predictable validation/CI workflows.

What Phase 1 is NOT

- It is not the dropout prediction model.
- It is not the counselor dashboard.
- It is not a one-off portal configuration.
- It is not a replacement for Phase 2.

What Phase 1 enables: Phase 2 gets a controlled, repeatable environment in which the application can run. The same foundation can later be adapted for other applications with similar infrastructure requirements.

8. Phase 1 — Detailed Components

Azure Storage

Provides cloud storage for application data, containers/files and artifacts. The supplied architecture specifically describes reusable storage-account patterns and dynamic provisioning of sub-resources.

Azure Linux VM / Compute

Provides the proposed compute environment for application and ML workloads.

Virtual Networking

Provides connectivity and network boundaries between cloud resources. Azure VNets/subnets are foundational networking constructs.

Security Baseline

Applies secure defaults at the infrastructure layer: reduce unnecessary public exposure, use secure transport where applicable, and avoid hardcoded credentials.

Azure Key Vault

Provides centralized secure storage for secrets such as passwords, connection strings and access keys. The application should retrieve secrets through an authorized identity/access path rather than storing them in source code.

Terraform State / Resource Organization

Terraform needs state to understand managed resources and compare desired configuration with deployed infrastructure. Resource groups organize Azure resources and lifecycle management.

Do not confuse configuration with secrets: normal environment configuration can be kept in configuration files; sensitive credentials belong in a secrets-management mechanism such as Key Vault.

9. Terraform Repository & Provisioning Workflow

Recommended repository structure

```
project/ ■■■ environments/ ■ ■■■ dev/ ■ ■■■ production/ ■■■ modules/ ■ ■■■ resource_group/
■ ■■■ storage/ ■ ■■■ virtual_machine/ ■ ■■■ networking/ ■ ■■■ security/ ■ ■■■ key_vault/
■■■ config/ ■ ■■■ *.yaml ■■■ .github/workflows/ ■■■ main.tf ■■■ variables.tf ■■■ outputs.tf
■■■ providers.tf ■■■ versions.tf ■■■ README.md
```

Provisioning workflow

Step	Purpose
terraform init	Initialize providers, modules and backend/state configuration.
terraform validate	Check configuration structure and syntax.
terraform plan	Preview intended infrastructure changes.
Review / CI	Run automated checks and team review before applying.
terraform apply	Provision or update the Azure environment.

The supplied Phase 1 architecture explicitly uses this predictable sequence. Microsoft and HashiCorp also document Terraform's init/plan/apply workflow for Azure.

Why this is better than clicking the portal: the environment becomes versionable, reviewable, repeatable and easier to reproduce.

10. Phase 2 — Student Retention Intelligence Platform

Phase 2 is the actual SIH application. The supplied architecture proposes predictive ML/XAI, prescriptive counseling and longitudinal delta tracking.

Module	Responsibility
Data Ingestion	Import/receive structured student metrics and validate the schema.
Risk Engine	Generate dropout-risk probability and risk band.
Explainability Engine	Generate student-level model contributor explanations.
Intervention Support	Map available drivers to transparent intervention categories.
Counselor Dashboard	Prioritize students and show risk, explanation and case information.
Case & Follow-up	Record selected intervention, notes and follow-up dates.
Longitudinal Tracking	Store risk history and observe change over time.
Analytics	Evaluate model performance, workload and intervention outcomes.

Example: a student appears at 78% risk; attendance decline and academic deterioration are strong model contributors; the counselor reviews the case and records an intervention; later the system records a new risk score such as 52%, then 31% if the student's trajectory improves.

11. Data & Feature Design

The supplied SIH analysis suggests starting with a public/de-identified dataset such as the UCI-based/Kaggle “Predict Students' Dropout and Academic Success” dataset. The exact dataset should be finalized after checking licensing, feature availability, class balance and suitability for our workflow.

Feature group	Examples	Purpose
Academic	grades, GPA, failed subjects, trend	Academic performance and deterioration.
Attendance	attendance %, recent trend	Early engagement warning.
Engagement	assignment completion, LMS activity if available	Participation/engagement signal.
Financial/admin	fee-related status if legitimately available	Potential support need; handle carefully.
Context/demographics	only where justified and authorized	Potential context, but higher fairness risk.

Feature engineering examples: recent attendance change, academic change over time, missed assignments in a time window, recent engagement trend and other features that are available before the prediction is made.

Do not invent features for accuracy. Every feature needs a defensible meaning, a reliable data source and a reason it would be available at prediction time.

12. Machine Learning Pipeline

The supplied SIH analysis recommends a baseline classification approach and stronger tree-based models. The supplied Phase 2 architecture mentions LightGBM/XGBoost. Our implementation should compare candidates rather than deciding the winner before evaluation.

Recommended sequence

1. Define the target label and prediction horizon clearly.
2. Clean and validate the dataset.
3. Perform EDA and inspect class balance.
4. Build a simple baseline such as Logistic Regression.
5. Compare Random Forest and XGBoost/LightGBM candidates.
6. Use a leakage-safe validation strategy.
7. Evaluate precision, recall, F1 and appropriate AUC metrics; inspect the confusion matrix.
8. Check probability calibration if risk probabilities are displayed.
9. Choose and document the final model/version.

Why accuracy is not enough: if the at-risk class is smaller, high accuracy can coexist with poor detection of the students we care about. The acceptable precision/recall trade-off should be chosen based on the intended human workflow and validation results.

13. Explainable AI with SHAP

SHAP is intended to show how individual features contributed to a model prediction. For a counselor, this is more actionable than seeing only a probability.

Example

The system predicts high risk. The student detail page shows that recent attendance decline is a strong contributor toward higher risk, while stable assessment performance contributes in the opposite direction. The counselor can inspect the underlying values and decide whether the explanation is consistent with the student's context.

UI should show

- Risk probability and risk band.
- Top contributing factors.
- Direction of contribution.
- Underlying feature values.
- Prediction timestamp and model version.
- A clear statement that the explanation supports human judgment.

Critical interpretation rule: SHAP contribution is not proof of causation. We should say “important model contributor” rather than “the reason this student will drop out.”

14. Counseling / Decision Layer

This layer turns model output into a human workflow. The system assists the counselor; it should not automatically decide a student's fate.

Observed pattern	Possible support category	Human role
Attendance deterioration	Attendance counseling / engagement check	Verify reason and choose action.
Academic decline	Academic support / mentoring / tutoring	Inspect subjects and context.
Fee/financial difficulty	Financial-aid / fee-support information	Verify eligibility and institutional policy.
Multiple signals	General counseling / case review	Investigate before selecting action.

Recommended design: transparent mappings first. Log what recommendation was shown, what the counselor selected, whether it was modified/rejected and the follow-up date.

Never automate a high-impact decision solely from the model score. The model is an early-warning and prioritization tool.

15. Longitudinal Tracking & Intervention Measurement

A one-time score describes one moment. Student retention is dynamic. The supplied architecture therefore proposes longitudinal delta tracking, where risk is recalculated and compared over time.

Time	Event	Example
T0	Initial prediction	78% risk
T0	Explanation	Attendance decline + academic deterioration
T0	Counselor action	Attendance-focused counseling
T1	Follow-up prediction	52% risk
T2	Later prediction	31% risk

What we store: prediction timestamp, model version, risk score, explanation summary, selected intervention, intervention date, follow-up date and subsequent observations.

Scientific caution: a lower risk after counseling is an outcome/association, not automatically proof that counseling caused the improvement. Other factors can change at the same time.

16. Counselor Product Modules

Risk Overview

- High/medium/low risk counts
- Newly elevated-risk students
- Students due for follow-up
- Risk trend summaries

Student Detail

- Current risk and probability
- Top SHAP contributors
- Recent feature values
- Historical risk trajectory
- Previous interventions

Case Management

- Open cases
- Counselor action
- Intervention status
- Follow-up date
- Notes/history

Intervention Analytics

- Interventions by category
- Follow-up completion
- Risk movement after intervention
- Model performance and alert volume

Optional: threshold-based email/notification can be added after the core prediction → explanation → counseling workflow is stable. The supplied SIH analysis treats alerts as optional.

17. End-to-End Technical Architecture

Layer	Main responsibility
Users	Authorized counselor/admin interacts with the system.
Frontend	React-based counselor interface.
Backend/API	Authentication, validation, business logic, prediction API and case workflow.
ML service	Loads approved model and returns prediction + explanation payload.
Storage	Student/application data and artifacts in Azure Storage or the finalized data store.
Secrets	Azure Key Vault for sensitive credentials/keys.
Compute	Azure Linux VM / selected Azure compute in the proposed architecture.
Networking	VNet/subnet/access controls and secure connectivity.
Infrastructure	Terraform modules and configuration provision the Azure foundation.
CI/CD	Automated infrastructure/application checks and controlled deployment.

Data path: *Student data → validation → feature preparation → ML prediction → SHAP explanation → intervention support → counselor action → persistence → later re-scoring.*

18. Recommended Technology Stack

Layer	Technology	Use
Cloud	Microsoft Azure	Cloud platform.
IaC	Terraform	Infrastructure as Code.
Config	YAML + Terraform variables	Environment-specific configuration.
Compute	Azure Linux VM	Application/ML runtime in proposed architecture.
Storage	Azure Storage	Data/artifact storage according to final design.
Secrets	Azure Key Vault	Secure credentials and secrets.
Backend	Python + FastAPI/Flask	APIs and integration.
ML	pandas + scikit-learn + XGBoost/LightGBM	Data + predictive modeling.
Explainability	SHAP	Local/global model explanations.
Frontend	React	Counselor UI.
Charts	Plotly or equivalent	Risk trends and analytics.
CI/CD	GitHub Actions or equivalent	Automated validation/deployment.
Version control	Git/GitHub	Collaboration and change history.

The supplied SIH analysis names Python, pandas, scikit-learn, XGBoost, SHAP, Flask/Streamlit and Plotly; the supplied architecture adds Azure/Terraform and proposes LightGBM/XGBoost for the predictive layer.

19. Development Roadmap

Stage	Work	Definition of done
1	Scope + architecture	Every teammate can explain the system.
2	Phase 1 cloud foundation	Terraform can provision/validate the development environment.
3	Data pipeline	Dataset and feature schema are reproducible.
4	Baseline ML	Baseline model + metrics documented.
5	Final ML candidate	XGBoost/LightGBM candidate evaluated and selected.
6	SHAP	Student-level explanation works.
7	Backend + UI	Risk and student-detail workflow works.
8	Counseling workflow	Intervention and follow-up can be recorded.
9	Longitudinal loop	New score updates risk trajectory.
10	Testing	Infrastructure, data, ML, security and UI tested.
11	Demo + docs	End-to-end demo is reproducible.

Team-lead rule: do not add advanced features until the core loop works end-to-end.

20. Team Roles

Area	Primary ownership	Everyone must understand
Cloud / DevOps	Terraform, Azure, networking, Key Vault, CI/CD	How Phase 2 uses Phase 1.
ML / Data	Dataset, EDA, features, models, evaluation, SHAP	What the application does with model output.
Backend / Integration	APIs, validation, persistence, model serving	ML pipeline + counselor workflow.
Frontend / Product	Counselor UI, risk views, explanations, charts	Meaning and limits of ML outputs.

For a two-member team: one member can take primary Cloud/DevOps ownership and the other primary ML/Application ownership, but both should pair on backend integration, security and final testing.

Working rules

- Document every major architecture decision.
- Do not commit secrets or student PII to Git.
- Review Terraform plans before applying infrastructure changes.
- Version the model and feature schema.
- Run an end-to-end demo regularly, even before the UI is polished.

21. Testing & Evaluation

Infrastructure

- Terraform format/validation checks.
- Plan review for unintended changes.
- Provision/destroy test in development.
- Network exposure and secret-handling checks.

Data

- Schema/type/range checks.
- Missing values and duplicates.
- Leakage checks.
- Feature availability at prediction time.

ML

- Precision, recall, F1 and suitable AUC metrics.
- Confusion matrix.
- Calibration if probabilities are shown.
- Performance across important subgroups where sample size permits.

Product

- Can the counselor find high-priority students?
- Can the counselor understand the model output?
- Can an intervention be recorded?
- Can a later risk trajectory be displayed?
- Can the team reproduce the cloud environment from code?

Success is not one accuracy number. The final evaluation should combine ML performance, explanation quality, workflow usability, infrastructure reproducibility and security/privacy checks.

22. Privacy, Fairness & Responsible AI

This is an educational decision-support system. Students should not be treated as labels generated by a model. The product must keep human review at the center.

Privacy

- Use public/de-identified data for development unless real institutional data is explicitly authorized.
- Minimize personally identifiable information.
- Separate identifiers from analytical features where possible.
- Use role-based access and secure secrets management.
- Do not put credentials in source code.

Fairness

- Inspect performance across relevant groups when data and ethics permit.
- Be careful with sensitive attributes and proxy variables.
- Do not use sensitive variables simply because they improve accuracy.
- Monitor false-positive/false-negative differences where appropriate.

Human-in-the-loop

The model prioritizes and explains. An authorized counselor reviews context and chooses the action. The system should never present a risk score as a final statement about a student's future.

23. MVP vs Advanced Scope

MVP — must work	Advanced — add later
Public/de-identified dataset	Authorized institutional integration
Data validation	Scheduled automated ingestion
Baseline + selected strong model	Advanced model comparison UI
Risk score/risk band	Advanced threshold optimization
SHAP explanation	Global explanation analytics
Counselor risk list	Advanced role hierarchy
Student detail page	Full case-management workflow
Transparent intervention categories	Advanced recommendation engine
Risk history	Advanced intervention-effect analytics
Terraform Azure foundation	More automated multi-environment releases

A complete MVP should demonstrate the full concept even if advanced automation is unfinished. One coherent closed loop is better than many disconnected features.

24. What Each Teammate Should Research

Topic	Learn	Expected mini-output
Terraform	Providers, resources, variables, modules, state, plan/apply	Provision one Azure resource safely.
Azure networking	VNet, subnet, NSG, public/private access	Simple network diagram.
Key Vault	Secrets, RBAC, identity, secure access	Secret retrieval demo.
Modules	Inputs, outputs, composition	Reusable storage/VM module.
YAML config	Schema and mapping to Terraform variables	One environment config.
CI/CD	Validation, plan, approvals, secure credentials	Infrastructure pipeline proof.
Dropout ML	Classification, imbalance, metrics	Evaluation report.
XGBoost/LightGBM	Boosting, tuning, feature importance	Model comparison.
SHAP	Local/global explanation	Student explanation demo.
Counseling workflow	Human-in-the-loop decision support	Intervention mapping.
Longitudinal data	Snapshots, events, deltas	Risk trajectory schema.
Security/privacy	PII, access, secrets, threat model	Security checklist.

Research rule: every technology you learn should end with a small working experiment and a short README explaining what you learned.

25. Questions the Team Must Be Able to Answer

Why not just build the ML model?

Because SIH25102 includes counseling. The product needs a workflow from early warning to human intervention and follow-up.

Why Phase 1?

Because a cloud application needs a consistent environment. Terraform makes infrastructure repeatable, reviewable and reusable.

Is Phase 1 separate from SIH25102?

No. It is the infrastructure foundation for the primary SIH25102 application.

Can other applications use Phase 1?

The modules can be reused or adapted for similar infrastructure requirements. We should describe it as reusable infrastructure, not a universal platform.

What data will we use?

Start with the public/de-identified dataset identified in the supplied SIH analysis. Real institutional data requires authorization and privacy controls.

What is actually differentiated?

The integrated architecture: reusable cloud foundation + explainable prediction + intervention workflow + longitudinal measurement.

Will AI counsel students automatically?

No. It provides decision support; an authorized counselor reviews and chooses the action.

What if the model is wrong?

We evaluate errors, expose explanations, use appropriate thresholds and keep human review.

How do we know an intervention worked?

We track later risk/data changes. We must not claim causality without a rigorous causal design.

What should the final demo show?

Provisioned environment → data ingestion → risk → SHAP → counselor action → follow-up → later risk trajectory.

26. Final Project Definition

We are building:

*A **Student Retention Intelligence Platform** for SIH25102 that uses explainable machine learning to identify students who may be at risk of dropout, explain important model contributors, support counselor intervention decisions, record follow-up actions, and observe risk changes over time.*

How we are building it:

1. Create a reusable Azure cloud foundation with Terraform.
2. Provision storage, compute, networking, security and secrets-management components through modular infrastructure code.
3. Use configuration-driven provisioning and validate changes before applying them.
4. Build a reproducible student-data and feature pipeline.
5. Train and evaluate baseline and stronger classification models.
6. Add SHAP explanations for student-level predictions.
7. Build counselor views around risk, explanation, intervention and follow-up.
8. Store risk history and intervention events for longitudinal tracking.
9. Test infrastructure, data, model, security and user workflow behavior.
10. Document the system so another teammate can reproduce and explain it.

Core loop: Predict → Explain → Decide → Intervene → Follow-up → Measure → Re-score

Phase 1: Cloud foundation. **Phase 2:** AI retention application. **Primary PS:** SIH25102.

27. Official Technical Research Links

Use primary documentation first. These are official references for the infrastructure technologies in our architecture.

Microsoft Learn — Terraform on Azure

<https://learn.microsoft.com/en-us/azure/developer/terraform/>

Terraform on Azure: providers, authentication, state, VMs, networking and other Azure services.

HashiCorp — Get Started with Azure

<https://developer.hashicorp.com/terraform/tutorials/azure-get-started>

Hands-on Terraform tutorials for Azure infrastructure, variables, changes and state.

Microsoft Learn — Terraform Fundamentals

<https://learn.microsoft.com/en-us/training/paths/terraform-fundamentals/>

Declarative infrastructure, modules, variables, outputs and repeatable deployment.

Microsoft Learn — Azure Virtual Network

<https://learn.microsoft.com/en-us/azure/virtual-network/>

VNets, routing, traffic filtering, peering and network security.

Microsoft Learn — Azure Key Vault

<https://learn.microsoft.com/en-us/azure/key-vault/general/>

Keys, secrets, certificates, access control and secure application integration.

Microsoft Learn — Key Vault Secrets

<https://learn.microsoft.com/en-us/azure/key-vault/secrets/>

Passwords, connection strings, access keys and secret-management scenarios.

Appendix — Source-to-Implementation Mapping

Source idea	Our implementation
SIH25102 dropout prediction + counseling	Risk engine + counselor decision-support workflow.
Attendance/marks/fee warning signals	Feature engineering using available, justified data.
Explainable prediction	SHAP-based local explanations.
Teacher/counselor dashboard	Risk overview + student detail + case workflow.
Optional alerts	Threshold notifications after core workflow.
Standardized IaC	Terraform modules and version-controlled infrastructure.
YAML-driven configuration	Environment-specific configuration separated from reusable modules.
Security baseline	Reduced exposure, secure transport, secrets management, identity-based access where supported.
Dynamic provisioning	Terraform iteration for repeated storage sub-resources.
Validation / CI	init → validate → plan → review/CI → apply.
Predictive ML / XAI	LightGBM/XGBoost candidates + SHAP.
Longitudinal tracking	Risk history + intervention events + re-scoring.

Source note: The SIH problem statement and supplied SIH25102 analysis, plus the supplied cloud/AI architecture, are the project-specific basis. Official Microsoft/HashiCorp links are external technical references. Where the source material does not prescribe an exact implementation choice, this guide treats the choice as a recommended team design rather than a mandated requirement.

End of team guide.